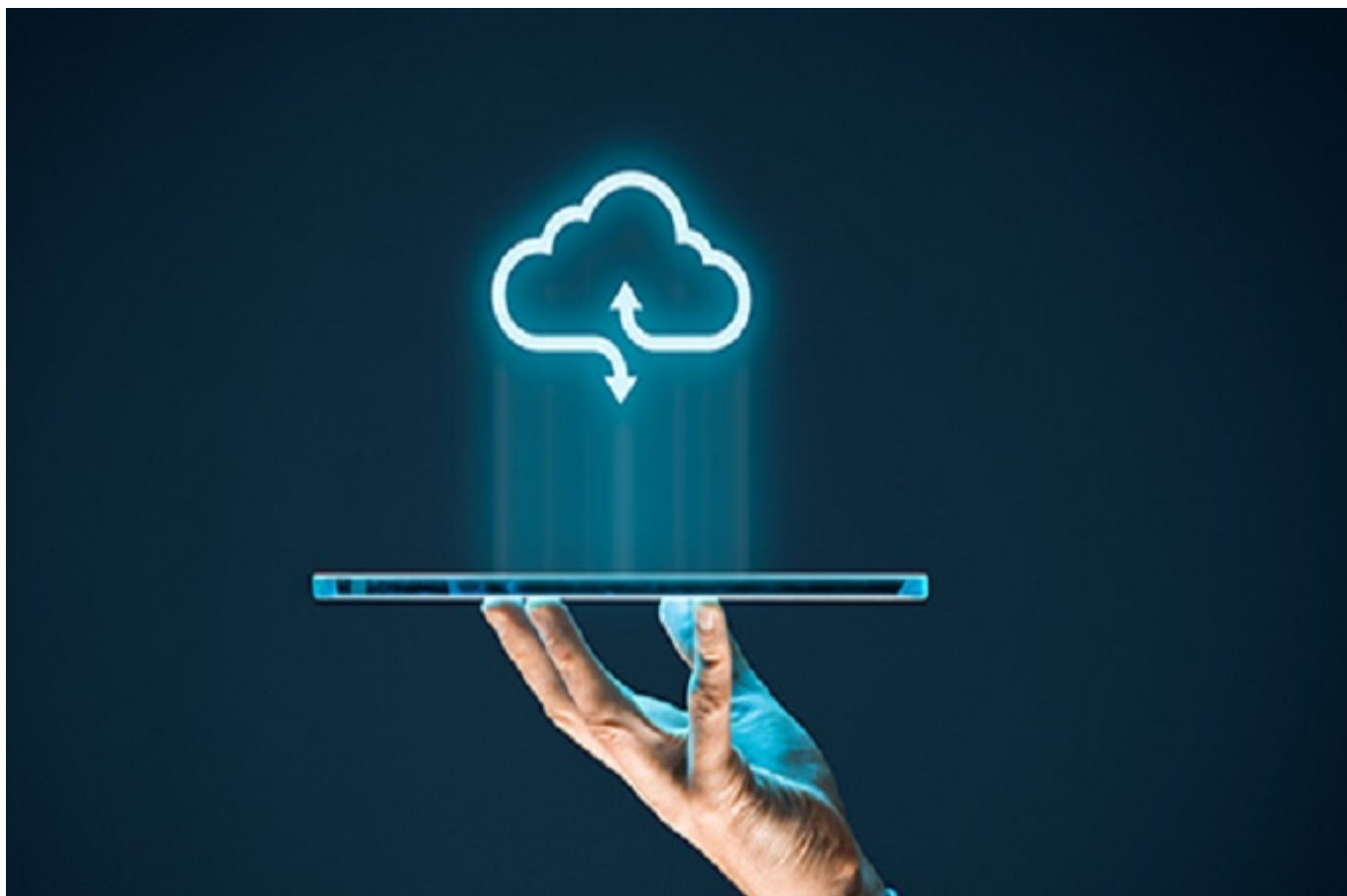




Cloud Computing

QP Code: SSC/Q8310

Version: 1.0

NSQF Level: 4.5

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
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Qualification Pack

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Qualification Pack

SSC/Q8310: Cloud Computing

Brief Job Description

Individuals at this job are responsible for designing and implementing information technology infrastructure in a public cloud environment and also the business and functionality for architecting infrastructure design.

Personal Attributes

Individuals must be able to communicate effectively. They must have strong interpersonal skills, analytical mindset and a zeal to learn new things.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8339: Work organization and management for cloud computing](#)
2. [SSC/N8341: Communication and interpersonal skills for cloud computing](#)
3. [SSC/N8342: Problem solving, innovation, and creativity](#)
4. [SSC/N8343: Cybersecurity](#)
5. [SSC/N8344: Reliability, scalability, and elasticity](#)
6. [SSC/N8345: Performance and optimization](#)
7. [SSC/N8346: Operational considerations](#)
8. [SSC/N8340: Sustainability.](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
Country	India
NSQF Level	4.5

Qualification Pack

Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ NIL
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) OR Pursuing 1st year of UG (continuing education) OR Previous relevant Qualification of NSQF Level (4) with 1-2 Years of experience relevant field
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-IT-01804-2024-V1-NASSCOM
NQR Version	1.0

Qualification Pack

SSC/N8339: Work organization and management for cloud computing

Description

Work organization and management for cloud computing

Scope

The scope covers the following :

- Work organization and management for cloud computing

Elements and Performance Criteria

Work organization and management

To be competent, the user/individual on the job must be able to:

- PC1.** identify common deployment models with public cloud providers and how those models can apply to organization-specific requirements.
- PC2.** identify opportunities and create migrations plans to phase-in public cloud deployments and reduce risks.
- PC3.** create highly available, scalable, and secure IT architectural designs specific to each application, taking into account compute, storage, networking, database management, and deployment requirements
- PC4.** take advantage of public cloud provider solutions to reduce operational burden associated with service deployments

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work organization and management</i>	20	30	-	-
PC1. identify common deployment models with public cloud providers and how those models can apply to organization-specific requirements.	6	7	-	-
PC2. identify opportunities and create migrations plans to phase-in public cloud deployments and reduce risks.	6	8	-	-
PC3. create highly available, scalable, and secure IT architectural designs specific to each application, taking into account compute, storage, networking, database management, and deployment requirements	5	7	-	-
PC4. take advantage of public cloud provider solutions to reduce operational burden associated with service deployments	3	8	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8339
NOS Name	Work organization and management for cloud computing
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8341: Communication and interpersonal skills for cloud computing

Description

Communication and interpersonal skills for cloud computing

Scope

The scope covers the following :

- Communication and interpersonal skills for cloud computing

Elements and Performance Criteria

Communication and interpersonal skills for cloud computing

To be competent, the user/individual on the job must be able to:

- PC1.** discover and document key requirements and how they relate to public cloud offerings.
- PC2.** discover and document technology-specific opportunities to leverage public cloud offerings
- PC3.** translate business goals and objectives into briefs, designs, and plans, and present such documents to stakeholders and management teams.
- PC4.** clearly map departmental and technology-specific requirements and goals to public cloud solutions
- PC5.** using project-specific migration plans, facilitate the implementation of an organizational transition to public cloud resources

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communication and interpersonal skills for cloud computing</i>	20	30	-	-
PC1. discover and document key requirements and how they relate to public cloud offerings.	5	6	-	-
PC2. discover and document technology-specific opportunities to leverage public cloud offerings	5	6	-	-
PC3. translate business goals and objectives into briefs, designs, and plans, and present such documents to stakeholders and management teams.	5	6	-	-
PC4. clearly map departmental and technology-specific requirements and goals to public cloud solutions	3	6	-	-
PC5. using project-specific migration plans, facilitate the implementation of an organizational transition to public cloud resources	2	6	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8341
NOS Name	Communication and interpersonal skills for cloud computing
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8342: Problem solving, innovation, and creativity

Description

Problem solving, innovation, and creativity

Scope

The scope covers the following :

- Problem solving, innovation, and creativity

Elements and Performance Criteria

Problem solving, innovation, and creativity

To be competent, the user/individual on the job must be able to:

- PC1.** evaluate, select, and implement foundational cloud computing services such as compute, network, and storage
- PC2.** evaluate, select, and implement advanced cloud computing services such as managed data services, caching services, and automated scaling and availability features.
- PC3.** evaluate, select, and implement various network-related technologies to infrastructure design such as network communication protocols, sub netting, NAT, DNS, VPN, broadcast networking, and dynamic routing protocols
- PC4.** automate infrastructure creation and modification through the use of scripting or programming, and the use of infrastructure templates

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Problem solving, innovation, and creativity</i>	35	40	-	-
PC1. evaluate, select, and implement foundational cloud computing services such as compute, network, and storage	9	10	-	-
PC2. evaluate, select, and implement advanced cloud computing services such as managed data services, caching services, and automated scaling and availability features.	8	10	-	-
PC3. evaluate, select, and implement various network-related technologies to infrastructure design such as network communication protocols, sub netting, NAT, DNS, VPN, broadcast networking, and dynamic routing protocols	9	10	-	-
PC4. automate infrastructure creation and modification through the use of scripting or programming, and the use of infrastructure templates	9	10	-	-
NOS Total	35	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8342
NOS Name	Problem solving, innovation, and creativity
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8343: Cybersecurity

Description

Cybersecurity

Scope

The scope covers the following :

- Cybersecurity

Elements and Performance Criteria

Cybersecurity

To be competent, the user/individual on the job must be able to:

- PC1.** design and implement authentication processes at departmental and organizational level, controlling access to public cloud administrative capabilities and system access.
- PC2.** develop policies and procedures for systems and application access to public cloud interfaces and services.
- PC3.** implement policies and procedures for auditing of public cloud activities and access.
- PC4.** create internal prescriptive guidance and requirements for procedures necessary to create, update, remove and access public cloud infrastructure and resources.
- PC5.** implement service and technology specific security controls on resources running within a public cloud environment as well as utilization of services provided by an IaaS vendor.
- PC6.** engage with business, development, and leadership staff to identify, recommend, and implement security best practices while ensuring an efficient user experience.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Cybersecurity</i>	50	50	-	-
PC1. design and implement authentication processes at departmental and organizational level, controlling access to public cloud administrative capabilities and system access.	5	5	-	-
PC2. develop policies and procedures for systems and application access to public cloud interfaces and services.	5	5	-	-
PC3. implement policies and procedures for auditing of public cloud activities and access.	10	10	-	-
PC4. create internal prescriptive guidance and requirements for procedures necessary to create, update, remove and access public cloud infrastructure and resources.	10	10	-	-
PC5. implement service and technology specific security controls on resources running within a public cloud environment as well as utilization of services provided by an IaaS vendor.	10	10	-	-
PC6. engage with business, development, and leadership staff to identify, recommend, and implement security best practices while ensuring an efficient user experience.	10	10	-	-
NOS Total	50	50	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8343
NOS Name	Cybersecurity
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8344: Reliability, scalability, and elasticity

Description

Reliability, scalability, and elasticity

Scope

The scope covers the following :

- Reliability, scalability, and elasticity

Elements and Performance Criteria

Reliability, scalability, and elasticity

To be competent, the user/individual on the job must be able to:

- PC1.** record, analyse, and interpret application, system, and network data to facilitate the recommendation of an appropriate architecture that sufficiently utilizes scalability and elasticity to meet the variable demands of internal and external users and systems.
- PC2.** implement different availability, scalability, and durability models in accordance with application and system design requirements
- PC3.** design availability models that meet the business requirements of an organization, taking into account allowed recovery time and allowable service interruption parameters
- PC4.** utilize public cloud services and features to aid the design and deployment of availability, durability, and scalability requirements.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Reliability, scalability, and elasticity</i>	30	45	-	-
PC1. record, analyse, and interpret application, system, and network data to facilitate the recommendation of an appropriate architecture that sufficiently utilizes scalability and elasticity to meet the variable demands of internal and external users and systems.	5	10	-	-
PC2. implement different availability, scalability, and durability models in accordance with application and system design requirements	5	10	-	-
PC3. design availability models that meet the business requirements of an organization, taking into account allowed recovery time and allowable service interruption parameters	13	10	-	-
PC4. utilize public cloud services and features to aid the design and deployment of availability, durability, and scalability requirements.	10	15	-	-
NOS Total	30	45	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8344
NOS Name	Reliability, scalability, and elasticity
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8345: Performance and optimization

Description

Performance and optimization

Scope

The scope covers the following :

- Performance and optimization

Elements and Performance Criteria

Performance and optimization

To be competent, the user/individual on the job must be able to:

- PC1.** analyse and interpret performance metrics from compute, storage, network, and application levels for use in public cloud infrastructure design objectives.
- PC2.** utilize performance tuning techniques and packages to ensure optimal resource utilization
- PC3.** implement microservice strategies to capitalize on technology advances in areas like container development.
- PC4.** pursue the decoupling of services to allow the separation of application components to facilitate a service-oriented architecture.
- PC5.** recommend and implement database and storage solutions that best fit the needs of an application.
- PC6.** implement serverless architecture

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Performance and optimization</i>	15	35	-	-
PC1. analyse and interpret performance metrics from compute, storage, network, and application levels for use in public cloud infrastructure design objectives.	3	5	-	-
PC2. utilize performance tuning techniques and packages to ensure optimal resource utilization	2	5	-	-
PC3. implement microservice strategies to capitalize on technology advances in areas like container development.	2	5	-	-
PC4. pursue the decoupling of services to allow the separation of application components to facilitate a service-oriented architecture.	3	5	-	-
PC5. recommend and implement database and storage solutions that best fit the needs of an application.	3	5	-	-
PC6. implement serverless architecture	2	10	-	-
NOS Total	15	35	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8345
NOS Name	Performance and optimization
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8346: Operational considerations

Description

Operational considerations

Scope

The scope covers the following :

- Operational considerations

Elements and Performance Criteria

Operational considerations

To be competent, the user/individual on the job must be able to:

- PC1.** maintain functionality and availability within the requirements of a given system
- PC2.** maintain functionality and availability within the requirements of a given system's applications
- PC3.** use the metrics of a given system, network, and applications to assess the durability, availability, and performance of the infrastructure
- PC4.** respond to cybersecurity, availability, and performance-related incidents by applying given requirements, protocols, and procedures

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operational considerations</i>	15	35	-	-
PC1. maintain functionality and availability within the requirements of a given system	4	9	-	-
PC2. maintain functionality and availability within the requirements of a given system's applications	4	8	-	-
PC3. use the metrics of a given system, network, and applications to assess the durability, availability, and performance of the infrastructure	3	9	-	-
PC4. respond to cybersecurity, availability, and performance-related incidents by applying given requirements, protocols, and procedures	4	9	-	-
NOS Total	15	35	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8346
NOS Name	Operational considerations
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8340: Sustainability.

Description

Sustainability

Scope

The scope covers the following :

- Sustainability

Elements and Performance Criteria

Sustainability

To be competent, the user/individual on the job must be able to:

- PC1.** architect cloud workloads to maximize sustainability and minimize environmental impact
- PC2.** measure the impact of given cloud workloads and model the future impact of workloads
- PC3.** identify and estimate the sources of impacts across the lifetime of given products
- PC4.** compare productive output of given products with their impacts, per work unit
- PC5.** create KPIs for given products' improved productivity, evaluate the options for meeting the KPIs, including their cost implications, and estimate their relative impact over time
- PC6.** research new and alternative hardware and software to support energy efficiency, and identify their implications for design
- PC7.** review the implications for sustainability of using managed services across a wide customer base
- PC8.** explain and illustrate to customers their options for sustainable solutions, their relative cost benefit and wider implications

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sustainability</i>	15	35	-	-
PC1. architect cloud workloads to maximize sustainability and minimize environmental impact	2	4	-	-
PC2. measure the impact of given cloud workloads and model the future impact of workloads	2	4	-	-
PC3. identify and estimate the sources of impacts across the lifetime of given products	2	4	-	-
PC4. compare productive output of given products with their impacts, per work unit	2	4	-	-
PC5. create KPIs for given products' improved productivity, evaluate the options for meeting the KPIs, including their cost implications, and estimate their relative impact over time	2	4	-	-
PC6. research new and alternative hardware and software to support energy efficiency, and identify their implications for design	1	5	-	-
PC7. review the implications for sustainability of using managed services across a wide customer base	2	5	-	-
PC8. explain and illustrate to customers their options for sustainable solutions, their relative cost benefit and wider implications	2	5	-	-
NOS Total	15	35	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8340
NOS Name	Sustainability.
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Carry out assessments under realistic work pressures found in the normal industry workplace (or simulated workplace).
- Ensure that the range of materials, equipment, and tools that learners use are current and of the type routinely found in the normal industry workplace (or simulated workplace) environments.
- Assess that all tasks and functions are completed in a way, and to a timescale that is acceptable in the normal industry workplace.
- Assign workplace (or simulated workplace) responsibilities that enable learners to meet the requirements of the NOS.
- Productivity levels must be checked to ensure that it reflects those that are found in the work situation being replicated.
- Provide situations that allow learners to interact with the range of personnel and contractors found in the normal industry workplace (or simulated workplace).

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8339.Work organization and management for cloud computing	20	30	0	0	50	10
SSC/N8341.Communication and interpersonal skills for cloud computing	20	30	0	0	50	10
SSC/N8342.Problem solving, innovation, and creativity	35	40	0	0	75	15
SSC/N8343.Cybersecurity	50	50	0	0	100	20
SSC/N8344.Reliability, scalability, and elasticity	30	45	0	0	75	15
SSC/N8345.Performance and optimization	15	35	0	0	50	10
SSC/N8346.Operational considerations	15	35	0	0	50	10
SSC/N8340.Sustainability.	15	35	0	0	50	10
Total	200	300	0	0	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.